

UICT — Uganda
Department of Computer Science

CSC 123

Database Systems Design I

Relational Databases & SQL

Ground to Mastery Study Notes

Bravehart
DCS Year 1, Semester 2
Academic Year 2025/2026

Assessment: Coursework 40% | Final Exam 60%

Assessment Breakdown

Component	Weight	Details
Continuous Assessment (Coursework)	40%	Tests, assignments, practicals
Final Examination	60%	Written final exam at semester end

Minimum pass mark: 50% overall

You must pass BOTH components to pass the course unit.

Unit 1: Introduction to Databases

A database is an organised collection of related data stored and managed so that it can be easily accessed, updated, and controlled. Databases replace traditional flat-file systems and eliminate problems such as data redundancy, inconsistency, and difficulty in access.

Term	Definition
Database	An organised collection of structured data stored electronically in a computer system
DBMS	Database Management System — software that manages databases (e.g., MySQL, Oracle, PostgreSQL)
Data	Raw, unprocessed facts and figures
Information	Processed data that is meaningful and useful for decision-making
Data redundancy	Unnecessary duplication of data across multiple files or tables
Data inconsistency	Different copies of the same data having different values — caused by redundancy
Data integrity	Accuracy, completeness, and consistency of data in a database

Advantages of DBMS over File Systems

- Reduced data redundancy — data is stored in one place
- Data consistency — all users access the same, up-to-date data
- Data security — access controls restrict unauthorised viewing or modification
- Data independence — application programs are insulated from data structure changes
- Concurrent access — multiple users can access and modify data simultaneously
- Backup and recovery — DBMS provides automated backup and crash recovery

EXAM TIP: Know the three levels of data abstraction: Physical level (how data is stored on disk), Logical level (what data is stored and relationships), View level (what the end user sees). This is the ANSI/SPARC three-schema architecture.

Unit 2: Entity-Relationship (ER) Modelling

The Entity-Relationship (ER) model is a conceptual tool used to design a database by defining the entities involved, their attributes, and the relationships between them. An ER diagram visually represents this model.

Concept	Definition
Entity	A real-world object or concept that can be distinctly identified (e.g., Student, Course)
Attribute	A property or characteristic of an entity (e.g., StudentID, Name, DOB)
Primary Key	An attribute (or set of attributes) that uniquely identifies each entity instance
Relationship	An association between two or more entities
Cardinality	Defines the numerical relationship between entities (1:1, 1:N, M:N)
Weak Entity	An entity that cannot be uniquely identified by its own attributes alone
Composite Attribute	An attribute that can be divided into smaller parts (e.g., Full Name = First + Last)
Multi-valued Attribute	An attribute that can have multiple values (e.g., Phone Numbers)

Cardinality Types

- One-to-One (1:1) — one instance of A relates to exactly one instance of B (e.g., Person : Passport)
- One-to-Many (1:N) — one instance of A relates to many instances of B (e.g., Department : Employees)
- Many-to-Many (M:N) — many instances of A relate to many instances of B (e.g., Students : Courses)

EXAM TIP: ER diagram symbols — Rectangle = Entity, Ellipse = Attribute, Diamond = Relationship, Double rectangle = Weak entity, Double ellipse = Multi-valued attribute, Dashed ellipse = Derived attribute. Know these for diagram questions.

Unit 3: Relational Database Model

The relational database model organises data into tables (relations) consisting of rows and columns. It was proposed by E.F. Codd in 1970 and remains the most widely used database model today.

Term	Definition
Relation (Table)	A two-dimensional structure with rows and columns for storing related data
Tuple (Row)	A single record in a table; represents one instance of the entity
Attribute (Column)	A named property of a relation; defines what data is stored
Domain	The set of valid values for an attribute (e.g., Age must be 0-150)
Primary Key	Uniquely identifies each tuple in a table; cannot be NULL or duplicate
Foreign Key	An attribute that references the primary key of another table; enforces referential integrity
Referential Integrity	Foreign key values must match an existing primary key value in the referenced table
Candidate Key	Any attribute (or combination) that could serve as a primary key

Properties of a Relation

- No two tuples (rows) can be identical
- Each attribute has a unique name within the relation
- All values in a column belong to the same domain
- The order of rows and columns does not matter
- Each cell contains exactly one atomic (indivisible) value

EXAM TIP: Understand the difference between Primary Key (uniquely identifies rows in the SAME table) and Foreign Key (references a Primary Key in ANOTHER table). A Foreign Key can have NULL values unless constrained.

Unit 4: Normalisation

Normalisation is the process of organising a database to reduce redundancy and improve data integrity. It involves decomposing a large table into smaller, well-structured tables based on functional dependencies.

Normal Form	Rule/Condition
1NF (First Normal Form)	All attributes must be atomic (no repeating groups or multi-valued attributes)
2NF (Second Normal Form)	Must be in 1NF; every non-key attribute must depend on the WHOLE primary key (eliminates partial dependency)
3NF (Third Normal Form)	Must be in 2NF; no non-key attribute depends on another non-key attribute (eliminates transitive dependency)
BCNF (Boyce-Codd Normal Form)	Stronger version of 3NF; every determinant must be a candidate key

Types of Dependencies

- Functional dependency: attribute B is functionally dependent on A if each value of A determines exactly one value of B ($A \rightarrow B$)
- Partial dependency: a non-key attribute depends on PART of a composite primary key (violates 2NF)
- Transitive dependency: a non-key attribute depends on another non-key attribute (violates 3NF)

EXAM TIP: For exam questions, identify violations step by step. Check for 1NF first (atomic values), then 2NF (no partial dependency on composite key), then 3NF (no transitive dependency). Most exam tables have 2NF or 3NF violations.

Unit 5: Structured Query Language (SQL) — DDL

SQL (Structured Query Language) is the standard language for creating, modifying, and querying relational databases. DDL (Data Definition Language) commands define the structure of the database.

DDL Command	Purpose
CREATE DATABASE	Creates a new database
CREATE TABLE	Creates a new table with defined columns and constraints
ALTER TABLE	Modifies an existing table (add/remove columns, change data types)
DROP TABLE	Permanently deletes a table and all its data
TRUNCATE TABLE	Removes all rows from a table but keeps the table structure
CREATE INDEX	Creates an index on a column to speed up query searches

Common SQL Constraints

- PRIMARY KEY — uniquely identifies each row; no NULLs allowed
- FOREIGN KEY — links two tables; enforces referential integrity
- NOT NULL — prevents a column from storing NULL values
- UNIQUE — ensures all values in a column are different
- CHECK — validates that column values satisfy a specified condition
- DEFAULT — provides a default value when no value is specified

EXAM TIP: Know the difference between DDL, DML, and DCL. DDL (CREATE, ALTER, DROP) — defines structure. DML (SELECT, INSERT, UPDATE, DELETE) — manipulates data. DCL (GRANT, REVOKE) — controls access permissions.

Unit 6: SQL — DML Queries

DML (Data Manipulation Language) commands are used to insert, retrieve, update, and delete data in tables. SELECT is the most powerful and frequently used SQL command.

DML Command	Purpose
SELECT	Retrieves data from one or more tables
INSERT INTO	Adds new rows of data into a table
UPDATE	Modifies existing data in a table
DELETE FROM	Removes specific rows from a table
WHERE	Filters rows based on a condition
ORDER BY	Sorts the result set (ASC = ascending, DESC = descending)
GROUP BY	Groups rows with the same values for aggregate functions
HAVING	Filters groups after GROUP BY (WHERE cannot be used with aggregate functions)

SQL Aggregate Functions

- COUNT() — returns the number of rows
- SUM() — returns the total of a numeric column
- AVG() — returns the average value
- MAX() — returns the largest value
- MIN() — returns the smallest value

JOIN Types

- INNER JOIN — returns rows with matching values in both tables
- LEFT JOIN — returns all rows from the left table, matching rows from right
- RIGHT JOIN — returns all rows from the right table, matching rows from left
- FULL OUTER JOIN — returns all rows when there is a match in either table

EXAM TIP: Use HAVING to filter aggregate results (e.g., HAVING COUNT(*) > 2), not WHERE. WHERE filters individual rows BEFORE grouping; HAVING filters groups AFTER. This distinction is a common exam question.

Unit 7: Database Administration & Security

Database administration involves managing the database system to ensure its performance, availability, security, and integrity. The Database Administrator (DBA) is responsible for these tasks.

Term	Definition
DBA	Database Administrator — person responsible for installing, configuring, and maintaining the DBMS
Backup	A copy of the database created to restore data after failure or loss
Recovery	The process of restoring a database to a consistent state after a failure
Transaction	A logical unit of work that must be completed fully or not at all
ACID Properties	Atomicity, Consistency, Isolation, Durability — properties of reliable database transactions
Concurrency control	Managing simultaneous access to the database to prevent conflicts
GRANT	SQL command that gives a user specific privileges (e.g., SELECT, INSERT)
REVOKE	SQL command that removes previously granted privileges from a user

ACID Properties Explained

- Atomicity — a transaction is all-or-nothing; if any part fails, the whole transaction is rolled back
- Consistency — a transaction must take the database from one valid state to another
- Isolation — concurrent transactions do not interfere with each other
- Durability — once committed, a transaction's changes are permanent even after a system crash

EXAM TIP: Know ACID by heart — it is almost always examined. Also know: COMMIT saves a transaction permanently; ROLLBACK undoes all changes made since the last COMMIT.